The coupled rejection sampler

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Introduction

The article

Our group worked on the article "The coupled rejection sampler" published by Adrien Corenflos and Simo Särkkä on March 11, 2022.

ArXiv : https://arxiv.org/abs/2201.09585

Introduction

Definition

Given a probability space $(\Omega, \mathcal{A}, \mathbb{P})$, X, Y random variables taking values in E and p, q two distributions on E.
A pair (X, Y) of random variables is said to be a coupling if $X \sim p$, $Y \sim q$.
If $\mathbb{P}(X = Y) > 0$ we say that the coupling is diagonal.

Remark : In the case of a maximal (diagonal) coupling,
$\mathbb{P}(X = Y) = \int_E \min(p(x), q(x)) d\mu(x)$

Remark : Generally for a coupling (X, Y), X and Y are NOT independent random variables ! (e.g Continuous case).

Goal : Consider different ways of constructing a diagonal coupling, be it maximal or not.

Why ?

Theoretical tool : Couplings of distributions p, q may be of interest to study distances between distributions as the value of $\mathbb{P}(X \neq Y)$ is closely linked to $d_{TV}(p, q)$.

MCMC Methods : Couplings may be of interest to measure the mixing time of a Markov Chain (e.g Gibbs sampling). They give an indication of the time it takes for the chain to reach its stationary state.

Other core applications : Optimal transport theory, Wasserstein distance etc...

Methods

The Thorisson Algorithm

The Thorisson algorithm:

The original Thorisson algorithm returns a maximal coupling (X, Y) such that $\mathbb{P}(X = Y)$ is maximized. The modified version introduces a parameter $C \in (0, 1]$ that controls the variance of the execution time.

```
Algorithm 1: Rejection Sampling

1. Function RejectionSampling(p, q, C):
2.    Sample X ~ p and U ~ Uniform(0, 1)
3.    if U < min(q(X) / p(X), C) then Y = X
4.    else:
5.        repeat:
6.            Sample Z ~ q and U ~ Uniform(0, 1)
7.            if U > min(1, C * p(Z) / q(Z)) then Y = Z and stop
8.    Return Y
```

Rejection sampling algorithm

Rejection sampling is a method used to generate samples according to a target density $g(x)$ from a density $f(x)$, under the condition that there exists a constant $M > 0$ such that $g(x) \leq Mf(x)$ for all x.

```
Algorithm 1: Rejection Sampling

1. Function RejectionSampling(f, g, M):
2.    Sample X ~ f
3.    Sample U ~ Uniform(0, 1)
4.    if U < g(X) / (M * f(X)) then accept X
5.    else: reject X and repeat
6.    Return X
```

Rejection Sampling Proof

The joint distribution of (X, U) is given by:

$$(X, U) \sim f(x) \cdot 1_{[0,1]}(u)$$

since X and U are sampled independently.
The acceptance region is:

$$\mathcal{A} = \{(x, u) \mid u < \frac{g(x)}{Mf(x)}\}$$

The conditional density given acceptance is then:

$$\tilde{g}(x) = \int_{(x,u)\in\mathcal{A}} f(x) \cdot 1_{[0,1]}(u)\, du = f(x) \cdot \int_0^{\frac{g(x)}{Mf(x)}} du = \frac{g(x)}{M}$$

Therefore, the accepted samples follow the distribution $g(x)$.

What Thorisson's Algorithm Has to Do with Rejection Sampling

In the second part of the algorithm, an implicit rejection sampling is performed:

- Target distribution: $q(z)$ (we want $Y \sim q$)
- Sampling density: $q(z)$ (self-sampling)
- Rejection criterion: accept Z if $U > \min\left(1, \frac{Cp(Z)}{q(Z)}\right)$

Validity of the Coupling:

We now show:

1. $X \sim p$: By construction.
2. $Y \sim q$: Let's decompose the pair (X, Y) into two cases:

- **Direct coupling:** $Y = X$
  The joint distribution of (X, U) is $(X, U) \sim p(x) \cdot 1_{[0,1]}(u)$, and the acceptance region is:

$$\mathcal{A} = \{(x, u) \mid u < \min\left(\frac{q(x)}{p(x)}, C\right)\}$$

  The portion of the density of Y conditioned on acceptance when $Y = X$ is:

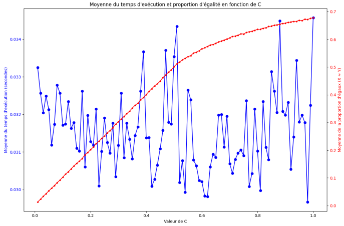
$$f_{=}(y) = p(y) \cdot \min\left(\frac{q(y)}{p(y)}, C\right) = \min\left(q(y), C \cdot p(y)\right)$$

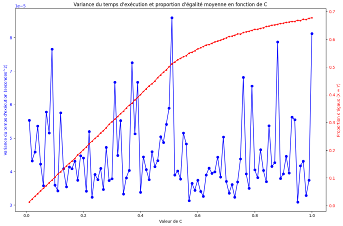
Validity of the Coupling

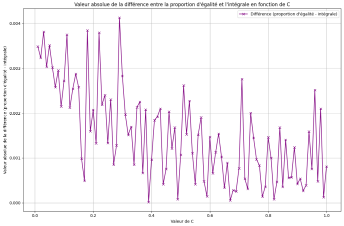
- **Resampling:** Y is generated separately.
  The joint distribution of (Z, U) is $(Z, U) \sim q(z) \cdot 1_{[0,1]}(u)$, and the acceptance region is:
  $$\mathcal{A} = \{(z, u) \mid u > \min\left(1, \frac{Cp(z)}{q(z)}\right)\}$$
  Thus, the portion of the density of Y conditioned on acceptance when $Y \neq X$ is:
  $$f_{\neq}(y) = q(y) \cdot \left(1 - \min\left(1, \frac{Cp(y)}{q(y)}\right)\right) = q(y) - \min(q(y), Cp(y))$$
  Finally, the density of Y is:
  $$f(y) = f_{=}(y) + f_{\neq}(y) = q(y)$$
- **Probability $\mathbb{P}(X = Y)$:**
  $$\mathbb{P}(X = Y) = \int_x \mathbb{P}(Y = x \mid X = x) \cdot p(x)\, dx = \int_x \min\left(\frac{q(x)}{p(x)}, C\right) \cdot p(x)\, dx$$
  $$\mathbb{P}(X = Y) = \int_x \min\left(q(x), Cp(x)\right)\, dx \neq C \int \min\left(p(t), q(t)\right)\, dt$$

Test with $\mathcal{N}(0,1)$ and $\mathcal{N}(0,2)$

Choice of C



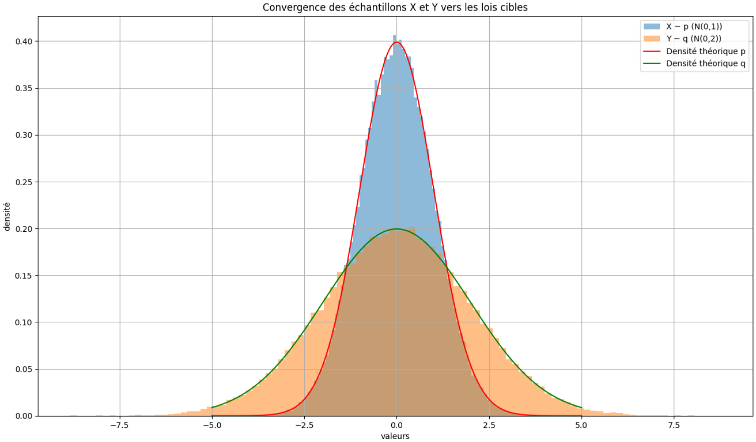




Choice of C = 1

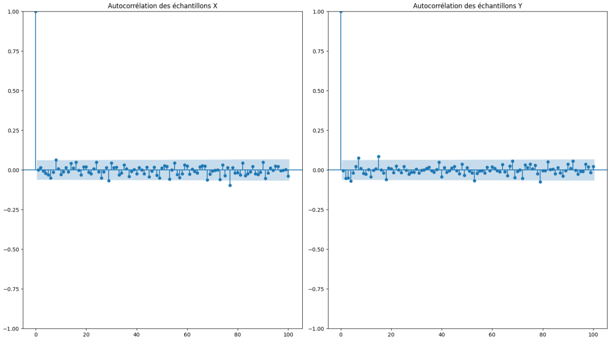
1. Equivalent to non modified Thorisson, returns a maximal coupling.
2. In practice execution time and variance of execution time are correct.
3. The absolute value of the difference between the integral and the average equality proportion is low.

Convergence of the samples



$\hat{\mathbb{P}}(X = Y) = 0.676$. Theoretical bound : $\int_E \min(Cp, q)d\mu = 0.677$

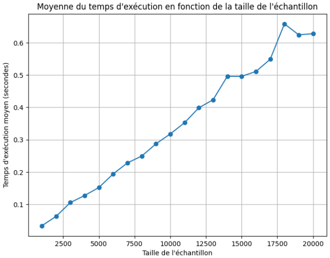
Autocorrelation test



Samples are generally uncorrelated

1. Slight variations of order ± 0.05 for some lag values (other than 0).
2. Deviation is not persistent over several lags.
3. Probably just statistical noise...

Performance analysis for different sample sizes



Linear increase in execution time

1. $O(n)$ complexity (in expected value).
2. Not really suitable for large sample sizes.

The Corenflos-Särkkä Approach

Validity and link to rejection sampling

Definition

We say that $\hat{p}$ (resp. $\hat{q}$) dominates p (resp.) q if there exists $1 \leq M(p, \hat{p}), M(q, \hat{q})$ such that :

$$p \leq M(p, \hat{p})\hat{p} \text{ and } q \leq M(q, \hat{q})\hat{q}.$$

We write $\hat{\Gamma} = (\hat{p}, \hat{q})$.

As in the traditional rejection sampling approach we consider $(X_1, Y_1) \sim \hat{\Gamma}$ two proposals and $U \sim \mathcal{U}([0, 1])$ independent. To get a diagonal coupling we may take $\hat{p} = \hat{q}$ same proposal distribution. In this case $\mathbb{P}(X = Y) = \mathbb{P}(\text{Both proposal are accepted}) > 0$.

The algorithm:

```
Algorithm 1: Rejection-Coupling of (p, q)

1. Function RejectionCoupling(Gamma_hat, p, q):
2.    Set AX = 0 and AY = 0    # Acceptance flags
3.    while AX == 0 and AY == 0 do:
4.        Sample (X1, Y1) ~ Gamma_hat
5.        Sample U ~ Uniform(0, 1)
6.        if U < p(X1) / (M(p, p_hat) * p_hat(X1)) then AX = 1
7.        if U < q(Y1) / (M(q, q_hat) * q_hat(Y1)) then AY = 1
8.    Sample (X2, Y2) ~ (p, q)
9.    Return:
          X = AX * X1 + (1 - AX) * X2
          Y = AY * Y1 + (1 - AY) * Y2
```

Proof :

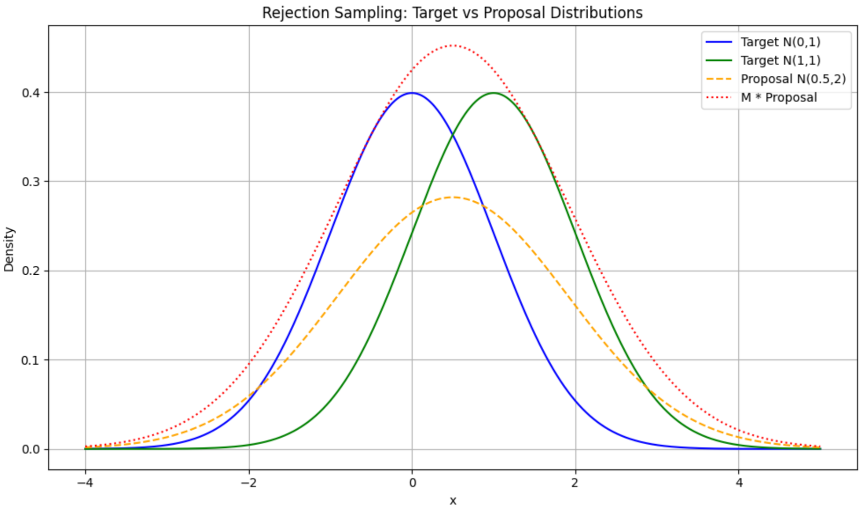
Coupling : Okay when we consider two possible case, both accepted or at least 1 rejected.

Diagonal : Suppose $\hat{\Gamma} \succeq (p,q)$ proposal and $\mathrm{Supp}(\hat{p}) \subset \mathrm{Supp}(p)$, same for q. Under the additional assumption that $\mathbb{P}(X_1 = Y_1) > 0, (X_1, Y_1) \sim \hat{\Gamma}$:

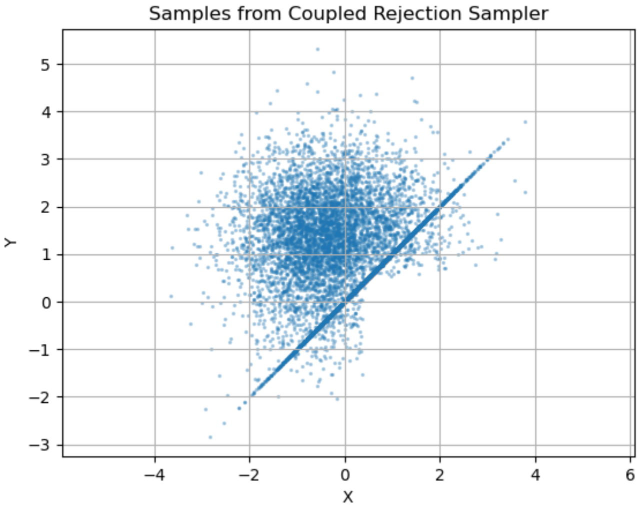
$$
\begin{aligned}
\mathbb{P}(X = Y) &\geq \mathbb{P}(X_1 = Y_1 \text{ and } A_X = A_Y = 1) \\
&= \mathbb{P}(A_X = 1, A_Y = 1 \mid X_1 = Y_1) \cdot \mathbb{P}(X_1 = Y_1) \\
&= \mathbb{P}(X_1 = Y_1) \int_E c(x) \min\left(\frac{p(x)}{M(p,\hat{p})\hat{p}(x)}, \frac{q(x)}{M(q,\hat{q})\hat{q}(x)} \right) d\mu(x) > 0.
\end{aligned}
$$

As c density of $X_1 \mid X_1 = Y_1$, as $\mathbb{P}(X_1 = Y_1) > 0$ there exists A of non-zero measure such that $c(A) > 0$, finally, for any $x \in A$, $\hat{p}(x), \hat{q}(x) > 0$ and the assumption gives us $p(x), q(x) > 0$.

Test case 1D, Proposal :



Test case 1D, Results :



$X \sim \mathcal{N}(0,1)$

$Y \sim \mathcal{N}(1,1)$

$n = 10000$

$\mathbb{P}(X = Y) \approx 0.4389$

Runtime is around 3s

Running time of the algorithms, a comparison I :

We know that in rejection sampling :

$$\tau \sim \mathcal{G}\left(\frac{1}{\mathbb{E}(\tau)}\right)$$

For the algorithm to finish we need $A_X = 1$ or $A_Y = 1$, but :

$$\tau_X \sim \mathcal{G}\left(\frac{1}{M(p,\hat{p})}\right),\ \tau_Y \sim \mathcal{G}\left(\frac{1}{M(q,\hat{q})}\right) \text{ and } \tau \leq \min(\tau_X, \tau_Y)$$

Thus :

$$\mathbb{E}(\tau) \leq \min(M(p,\hat{p}), M(q,\hat{q}))$$
$$\mathrm{Var}(\tau) <= \min(M(p,\hat{p}), M(q,\hat{q}))^2 - \min(M(p,\hat{p}), M(q,\hat{q}))$$

Running time of the algorithms, a comparison II :

The total variation distance d_{TV} is a measure of distance between measures. The theoretical bounds on couplings are functions of this distance.

The non modified Thorisson algorithm gives a maximal diagonal coupling, that is $X \sim p, Y \sim q$ and $\mathbb{P}(X = Y) = 1 - d_{TV}(p, q)$ (theoretical bound).

However depending on the distributions p, q, $\mathrm{Var}(\tau) \to +\infty$ when $d_{TV}(p, q) \to 0$.

Hence the approach by Corenflos and Särkkä gives us a bounded runtime variance at the cost of the optimality of the diagonal coupling.

The curse of dimensionality, improving the algorithm for high dimensions :

$\mathbb{P}(\text{Accept sample}) = \mathbb{P}\left(U \leq \frac{f(x)}{Mg(x)}\right) = \frac{1}{M}$

The main issue when sampling in increasing dimension (X_i's $\in \mathbb{R}^d$, $d \to +\infty$) :

Proposal $g(x) = \prod_{i=1}^{d} g_0(x_i)$, sample from $f(x) = \prod_{i=1}^{d} f_0(x_i)$, $\sup_x \frac{f_0(x)}{g_0(x)} = k_0 > 1$

Then $\sup_x \frac{f(x)}{f(x)} \leq k_0^d$ so $\mathbb{P}(\text{Accept sample}) = \frac{1}{k_0^d}$ decreases exponentially with dimension.

We want a more robust approach to do simulations in high dimensional spaces in reasonable time.

The ensemble method, intuition :

Instead of sampling just 1 point (low probability acceptance), sample N points from $\hat{\Gamma} = (\hat{p}, \hat{q})$ (often $\hat{p} = \hat{q}$).

For each point, calculate how good of a candidate it is for p and q : $W_i^X = \frac{p(X_i)}{\hat{p}(X_i)}$, same for Y_i's. Normalize those weights, the higher the value of w_i^X, the best candidate for p is X_i (same for Y_i).

We know have two distributions $\left(w_1^X, w_2^X, ..., w_N^X\right)$ and $\left(w_1^Y, w_2^Y, ..., w_N^Y\right)$ on the samples decribing how good they are for p and q respectively.

Compute the maximal coupling of those two distributions to maximise $\mathbb{P}(I = J)$ where I, J are indexes drawn. This maximises $\mathbb{P}(X = Y)$!

Adjust the final weights and accept samples if $U \leq \frac{\hat{Z}}{\bar{Z}}$, when $N \to +\infty$, proba of acceptance tends to the one of classical rejection.

Application to the d-dimensional Gaussian case :

Coupling d-dimensional Gaussians :

Coupling two Gaussians $\mathcal{N}(\mu_p, \Sigma_p)$ and $\mathcal{N}(\mu_q, \Sigma_q)$ with $\mu_p \neq \mu_q, \Sigma_p \neq \Sigma_q$ is an important special case.

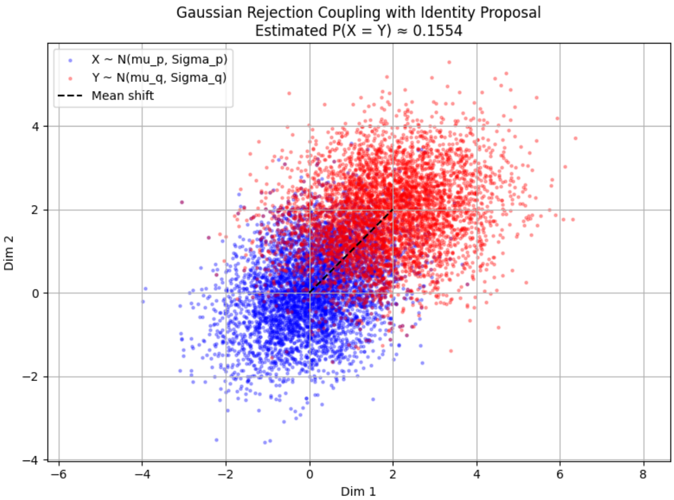
In this particular case, the author propose a general method to construct a proposal of Gaussians sharing the same covariance matrix $\hat{\Sigma}$.

In this particular case we also have explicit lower and upper bounds on $\mathbb{P}(X = Y)$.

Approach : $\hat{\Sigma}$ needs to verify $\hat{\Sigma}^{-1} \preceq \Sigma_p^{-1}, \hat{\Sigma}^{-1} \preceq \Sigma_q^{-1}$. Then with $M(p, \hat{p}) = \frac{\det(2\pi\hat{\Sigma})^{1/2}}{\det(2\pi\Sigma_p))^{1/2}}$ and $M(q, \hat{q}) = \frac{\det(2\pi\hat{\Sigma})^{1/2}}{\det(2\pi\Sigma_q)^{1/2}}$, algorithm 1 gives us the suitable coupling.

The proof is a direct consequence of rejection sampling for coupling.

Applications to two gaussians :



$$X \sim \mathcal{N}\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & 0.3\\0.3 & 1\end{pmatrix}\right)$$

$$Y \sim \mathcal{N}\left(\begin{pmatrix}2\\2\end{pmatrix}, \begin{pmatrix}1.5 & 0.2\\0.2 & 1\end{pmatrix}\right)$$

$n = 10000$

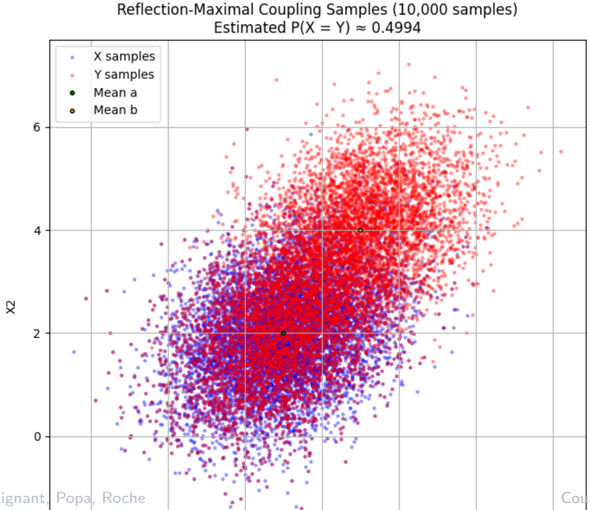
$\mathbb{P}(X = Y) \approx 0.155$

Runtime is around 5s

$\hat{\Sigma} = \max(\sigma(\Sigma_p), \sigma(\Sigma_q)) I_d$

Compare to same cov matrix

Applications to two gaussians, comparison (same cov matrix) :



$$X \sim \mathcal{N}\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}2 & 0.5\\0.5 & 1\end{pmatrix}\right)$$

$$Y \sim \mathcal{N}\left(\begin{pmatrix}2\\2\end{pmatrix}, \begin{pmatrix}2 & 0.5\\0.5 & 1\end{pmatrix}\right)$$

$n = 10000$

$\mathbb{P}(X = Y) \approx 0.499$

Runtime is around 7s

Maximal coupling here

Out of scope of paper...

Thank you!

References

- A. Corenflos, S. Särkkä. The Coupled Rejection Sampler, Arxiv preprint, 2022.
- Gerber, M. and Lee, A. (2021). Discussion on the paper "Unbiased Markov chain Monte Carlo methods with couplings" by Jacob, O'Leary and Atchad'e. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 82(3):584–585.